

# Config Register Reference — AXI → DFI

One channel, DDR5. Every programmable config register (not runtime state). **dom**: cfg=APB/config-clk, mc=mc\_clk(=dfi\_clk), ctrl=DFI control-clk, dev=DDR device. **src**: APB=host bus, MR=DDR mode-reg (via MRR / vendor-set), PHY=PHY-provided, FSM=freq-change FSM. All widths parametric.

## 1 Geometry / topology (static, boot)

| register  | width   | dom | src    | read by    | update when | function                                          |
|-----------|---------|-----|--------|------------|-------------|---------------------------------------------------|
| N_RANKS   | \$clog2 | cfg | APB    | all blocks | boot only   | rank count (drives CS/CKE/ODT widths, cell depth) |
| N_BG      | 3       | cfg | APB    | decode/arb | boot        | bank groups (8)                                   |
| N_BANKS   | derived | cfg | –      | cells/arb  | boot        | = N_BG×4                                          |
| BL        | 2       | cfg | APB/MR | data path  | boot        | burst length (16/32) -> BL/2 = dqFree             |
| DQ_WIDTH  | cfg     | cfg | APB    | data path  | boot        | sub-channel DQ (32) -> 2·DQ per phase             |
| ROB_DEPTH | cfg     | cfg | APB    | write ROB  | boot        | outstanding write reqs                            |
| GEAR_MAX  | 2       | cfg | APB    | phaser/DFI | boot        | max mc:CK ratio (sizes phase ports)               |

## 2 Address map (static)

| register                        | width | dom | src | read by          | update when | function                                                           |
|---------------------------------|-------|-----|-----|------------------|-------------|--------------------------------------------------------------------|
| amap_{rank,bg,bank,row,col}_pos | few×5 | mc  | APB | ADEC field-slice | boot        | bit positions of each field in daddr (pure slice, no hash MC-side) |

## 3 Timing constants (per-scope const banks; DVFS-reloadable)

| register                                                     | width    | dom | src          | read by       | update when        | function                                                   |
|--------------------------------------------------------------|----------|-----|--------------|---------------|--------------------|------------------------------------------------------------|
| BANK_CONST: tRC tRCD tRAS<br>tRTP tRP WR_TAIL                | 6×8      | mc  | APB /<br>FSM | bank adders   | init + freq-change | ACT/RD/WR/PRE bank deadlines (CK).<br>WR_TAIL=CWL+BL/2+tWR |
| BG_CONST: tRRD_L tCCD_L<br>tCCD_L_WR                         | 3×8      | mc  | APB /<br>FSM | BG adders     | init + freq-change | same-BG spacing                                            |
| RANK_CONST: tRRD_S tFAW<br>tRTW WTR_TAIL_L/S tPPD<br>tRTRS   | 7×8      | mc  | APB /<br>FSM | rank counters | init + freq-change | cross-BG ACT, 4-ACT window, turnaround, PPD, rank-hop      |
| CH_CONST: BL_HALF                                            | 8        | mc  | APB          | dqFree ctr    | init + freq-change | DQ occupancy (BL/2)                                        |
| MAINT_CONST: tRFCsb tRFC1<br>tRFM tZQCAL tZQLAT tMRD<br>tMOD | 7×(8-12) | mc  | APB /<br>FSM | maint gates   | init + freq-change | refresh/RFM/ZQ/MRW durations                               |

## 4 PHY interface latencies (PHY-provided; DVFS-reloadable)

| register    | width | dom | src | read by   | update when                 | function                                 |
|-------------|-------|-----|-----|-----------|-----------------------------|------------------------------------------|
| tphy_wrlat  | few   | mc  | PHY | WL_LAUNCH | init + freq-change + phyupd | WR cmd -> dfi_wrdata_en                  |
| tphy_wrdata | few   | mc  | PHY | WL_LAUNCH | init + freq-change          | dfi_wrdata_en -> dfi_wrdata              |
| tphy_rdlat  | few   | mc  | PHY | RD_CAP    | init + freq-change + phyupd | RD cmd -> dfi_rddata_en / capture window |

## 5 Gear / frequency (freq-change FSM)

| register            | width | dom  | src | read by          | update when        | function                        |
|---------------------|-------|------|-----|------------------|--------------------|---------------------------------|
| gear                | 2     | mc   | FSM | GC, all counters | DVFS (freq-change) | mc:CK ratio (1/2/4); GC += gear |
| dfi_cmd_freq_ratio  | 2     | ctrl | FSM | PHY              | DVFS               | command freq ratio (out on DFI) |
| dfi_data_freq_ratio | 2     | ctrl | FSM | PHY              | DVFS               | data freq ratio                 |

| register  | width | dom  | src         | read by  | update when     | function                           |
|-----------|-------|------|-------------|----------|-----------------|------------------------------------|
| freq_fsp  | 1-2   | ctrl | FSM         | PHY / MR | DVFS            | frequency set-point (which MR set) |
| frequency | W     | ctrl | FSM         | PHY      | DVFS            | target freq code                   |
| rr_shift  | 2     | mc   | FSM/MR-poll | refi_cnt | MR4 poll (temp) | tREFI » rr_shift (1x/2x/4x)        |

## 6 Refresh / RFM config

| register       | width | dom | src            | read by         | update when        | function                                |
|----------------|-------|-----|----------------|-----------------|--------------------|-----------------------------------------|
| tREFI          | 14    | mc  | APB            | refi_cnt reload | init + freq-change | refresh interval (CK)                   |
| debt_force_thr | 3     | mc  | APB            | refresh         | boot               | force at debt==7 (one guard before cap) |
| debt_cap       | 4     | mc  | APB            | refresh         | boot               | JEDEC postpone max (8)                  |
| RAAIMT         | 4     | dev | MR<br>(vendor) | RFM             | init (MRR read)    | RAA initial thr + decrement quantum     |
| RAAMMT         | 3     | dev | MR<br>(vendor) | RFM             | init (MRR read)    | RAA max thr (= k×RAAIMT, hard cap)      |
| raa_dec_mode   | 2     | mc  | APB/MR         | RFM             | init               | RAA decrement = RAAIMT or 0.5×          |
| refresh_mode   | 2     | mc  | APB/MR         | refresh         | init/temp          | 1x/2x/4x (FGR) refresh rate             |

## 7 DDR mode-register shadow (MC keeps copy; MRW-programmed; DVFS-reprogram)

| register                                                                          | width | dom | src             | read by    | update when        | function                                                                             |
|-----------------------------------------------------------------------------------|-------|-----|-----------------|------------|--------------------|--------------------------------------------------------------------------------------|
| MR_shadow: CL CWL BL_mode<br>RTT_WR/NOM/PARK DBI_rd/wr<br>CRC_rd/wr ODT VrefCA/DQ | many  | dev | APB<br>(values) | MRW engine | init + freq-change | device settings; re-sent via MRW on freq/temp change;<br>CL/CWL feed the tail consts |

## 8 Feature enables (static)

| register                       | width | dom | src | read by      | update when | function                                       |
|--------------------------------|-------|-----|-----|--------------|-------------|------------------------------------------------|
| ecc_en                         | 1     | mc  | APB | WR/RD path   | boot        | controller ECC (RMW path) vs on-die-only (DM)  |
| dbi_en / crc_en /<br>parity_en | 1 ea  | mc  | APB | data/CA path | boot        | DBI, link-CRC, CA-parity enables               |
| page_policy                    | 1     | mc  | APB | bank FSM     | boot        | open- vs closed-page                           |
| ap_policy                      | 2     | mc  | APB | CAS path     | boot        | auto-precharge (RDA/WRA) policy incl. drain AP |
| 2n_mode / cs_geardown          | 1 ea  | mc  | APB | DFI encode   | boot/init   | DDR5 2N / CS-geardown                          |

## 9 Scheduler tunables (static)

| register        | width | dom | src | read by         | update when | function                                |
|-----------------|-------|-----|-----|-----------------|-------------|-----------------------------------------|
| age_cap         | 4     | mc  | APB | CMD_ARB weights | boot        | starvation age-cap (fences both dirs)   |
| wlr_batch_len   | few   | mc  | APB | WLR arb         | boot        | R/W batch length (adaptive base)        |
| lookahead_depth | few   | mc  | APB | lookahead       | boot        | bypass window entries (4)               |
| bg_rotate_pref  | 1     | mc  | APB | BG_ARB          | boot        | diff-BG CAS preference (tCCD_S packing) |

## 10 ODT config

| register                       | width | dom | src     | read by    | update when        | function                                    |
|--------------------------------|-------|-----|---------|------------|--------------------|---------------------------------------------|
| RTT_WR / RTT_NOM /<br>RTT_PARK | 3×few | dev | APB/MR  | ODT gen    | init (MRW)         | termination values                          |
| odt_latency                    | few   | mc  | APB/PHY | ODT window | init + freq-change | ODT assert offset around write (WL-aligned) |

**Update cadence summary:** **boot-once** = geometry, addr-map, feature enables, scheduler tunables, thresholds. **DVFS / freq-change (quiesced)** = all timing consts, tphy\_\*, gear/freq-ratio, MR shadow (re-MRW), ODT. **MR-poll (runtime)** = rr\_shift / refresh\_mode (from MR4 temp). **init (MRR read)** = RAAIMT/RAAMMT (vendor). **phyupd (PHY-initiated)** = PHY may nudge tphy\_\* / its own delays. Config is written by APB (host) or the freq-FSM; never by traffic. All timing regs stay flops (freq-FSM writable) so they never constant-fold.